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Great — I'll create an 8–10 slide presentation (bullet points + code snippets) in Hinglish, ready for your intern talk. I'll produce slide titles, slide text (what to show), and speaker notes (exact lines you can say). Each slide will include one short code snippet or plot suggestion from the notebook and a simple explanation. I'll keep language mostly Hindi (Roman script) mixed with English technical terms — easy to read and present.

I'll start now — here are the slides:

Slide 1 — Title

- Show: "Gemstone Price Prediction" and your name (Nandini Katre), role: Intern, college: MCA (Nagpur).
- Speaker notes: "Namaste, main Nandini Katre hoon. Aaj main ek gemstone price prediction project present karungi — iska goal hai gem/diamond ke features se price predict karna."^[1]

Slide 2 — Problem statement & Dataset

- Show bullets:
 - Problem: Predict gemstone price (regression).
 - Key features: carat, cut, color, clarity, depth, table, x, y, z, price (target).
 - Dataset: loaded into pandas DataFrame (rows $\approx$ large number).
- Code snippet to show:
 - `df.head()` # show first 5 rows
- Speaker notes: "Yeh dataset pandas mein read kiya gaya; price ko target maana gaya aur baki columns features hain."^[1]

Slide 3 — Quick EDA (what we checked)

- Show bullets:
 - `df.describe()` to see ranges (carat, price, dimensions).
 - `df.nunique()` and `value_counts()` for categorical checks (cut: Ideal, Premium, ...).
 - Plots: `countplot(cut)` and price distribution (histogram).
- Code snippet to show:
 - `df.cut.value_counts()`
- Speaker notes: "Isse pata chala ki 'Ideal' cut sabse zyada hai; EDA se imbalance aur ranges samajhte hain."^[1]

Slide 4 – Data cleaning steps

- Show bullets:
 - Drop 'Unnamed: 0' extra column.
 - Check & fix zero/invalid x,y,z (remove or impute).
 - Encode categorical (cut,color,clarity).
- Code snippet to show:
 - `df.drop('Unnamed: 0', axis=1, inplace=True)`
- Speaker notes: "CSV se kabhi kabhi extra index column aata hai — usko drop kiya. Dimensions mein zeros check karna zaroori hota."^[1]

Slide 5 – Feature engineering & scaling

- Show bullets:
 - Create X (features) and y (price).
 - Possible derived feature: volume = xyz.
 - Scale numeric features (MinMax/Standard) so magnitudes similar.
- Code snippet to show:
 - `X = df.drop('price', axis=1); y = df['price']`
 - `from sklearn.preprocessing import MinMaxScaler`
- Speaker notes: "Scaling se model ko numeric feature magnitude ki wajah se bias nahi hota; volume ek acha engineered feature ho sakta hai."^[1]

Slide 6 – Models trained & how evaluated

- Show a small table (3 rows):
 - Decision Tree — MSE: (example) 1,608,550 — R2: 0.875^[1]
 - Random Forest — MSE: 233,142 — R2: 0.9818^[1]
 - XGBoost — MSE: 232,278 — R2: 0.9819^[1]
- Speaker notes: "Lower MSE and higher R2 better hote hain. Random Forest aur XGBoost sabse achhe nikle; dono non-linear patterns pakadte hain."^[1]

Slide 7 – Pipeline & training

- Show bullets:
 - Built sklearn Pipeline: preprocessing (scaler + encoder) + model.
 - Train with `train_test_split` (80/20).
- Code snippet to show:
 - `from sklearn.pipeline import Pipeline`
`pipeline = Pipeline([('scaler', MinMaxScaler()), ('model', RandomForestRegressor())])`
`pipeline.fit(X_train, y_train)`

- Speaker notes: "Pipeline ek hi object mein preprocessing aur model chain karta hai; deployment easy ho jata hai."^[1]

Slide 8 — Save model & demo predict

- Show bullets:
 - Save pipeline with pickle: `opengemstones.pkl`.
 - Demo code to load & predict one sample.
- Code snippet to show:
 - `import pickle`
`with open('opengemstones.pkl','wb') as f: pickle.dump(pipeline, f)`

Load & predict

```
with open('opengemstones.pkl','rb') as f: model = pickle.load(f)
sample =
{'carat':0.7,'cut':'Premium','color':'G','clarity':'VS2','depth':61.5,'table':57,'x':5.7,'y':5.75,'z':3.5
5}
pred = model.predict(pd.DataFrame([sample]))
```

- Speaker notes: "Yeh demo dikhaega ke ek sample input dene par model price predict karta hai. Presentation mein ek short live demo ya screenshot dikha sakti ho."^[1]

Slide 9 — Findings & Feature importance

- Show bullets:
 - Most important features: typically carat, clarity, color (explainable via model importances).
 - Random Forest/XGBoost give `feature_importances_` to explain.
- Code snippet to show:
 - `print(model.named_steps['model'].feature_importances_)`
- Speaker notes: "Agar time ho toh feature importances ya SHAP use karke explainability dikha sakti ho — hiring managers ko pasand aata hai."^[1]

Slide 10 — Next steps & improvements

- Show bullets:
 - Hyperparameter tuning (GridSearchCV / RandomizedSearchCV).
 - Cross-validation and outlier removal.
 - Deploy as Flask app or simple Streamlit for demo.
- Speaker notes: "Yeh improvements se model aur robust aur production-ready banega. Main agar chahu toh yeh bhi implement kar sakti hoon."^[1]

Extras — Slide design & delivery tips

- Keep each slide simple: 3–5 bullets, one small code box or plot.

- Time: 8–10 minutes presentation + 2–3 minutes Q&A.
- When asked technical question, say: "We used RandomForest for robustness; could try XGBoost hyper-tuning next."
- Have one backup slide with metrics and train/test split details (random_state used etc.).^[1]

I can now:

- Format these slides into a ready-to-download PowerPoint (pptx) file with the text and code snippets placed on slides, OR
- Provide the same content as plain text slide-by-slide (so you can paste into Google Slides / PowerPoint).

Which format do you want? (A) PowerPoint file, (B) Plain text slides to paste.

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1. gemstone.ipynb
2. <https://www.facebook.com/OptionALtd/>
3. <https://www.option-a.org/en/faq/>
4. <https://optionalpha.com/>
5. <https://www.option-a.org/en/>
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7. <https://play.google.com/store/apps/details?id=com.gmail.rconsumption.mainprojectcopycopy1&hl=eu>
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